

Lecture Week 2: TRANSCENDENTAL FUNCTIONS (REVIEW of HIGH SCHOOL MATHEMATICS)

A Library of Transcendental Functions.

2.1. Exponential Functions

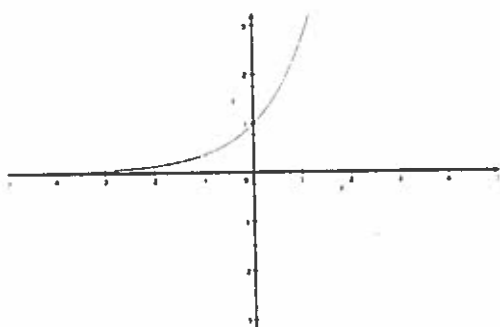
$$y = a^x \quad \text{where } a \in \mathbb{R}$$

$$\text{Dom } f(x) := \{x \in \mathbb{R}\}$$

$$\text{Ran } f(x) := \{y \in \mathbb{R} | y > 0\}$$

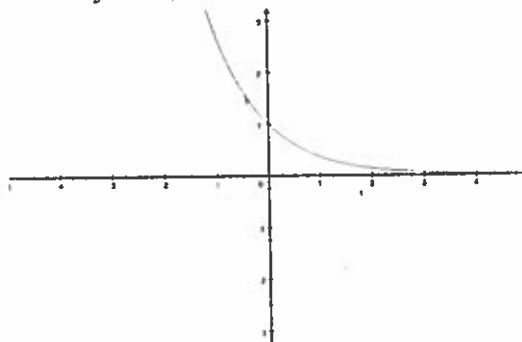
a) Exponential Growth Functions.

$$y = a^x, a \in \mathbb{R}, a > 1$$



b) Exponential Decay Functions

$$y = b^x, a \in \mathbb{R}, b < 1$$



2.2. Inverse Exponential Function (Logarithmic Function)

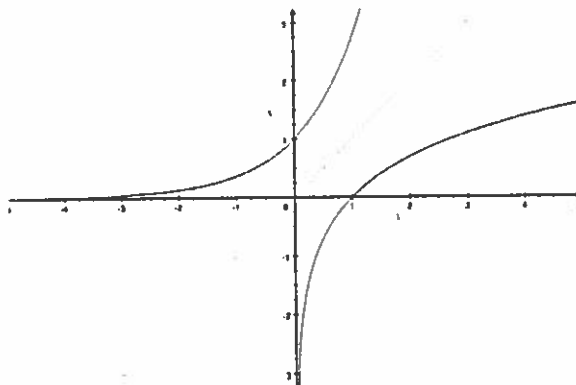
For a fixed $a > 0$ we defined the exponential function

$$f(x) = a^x \quad \text{with } \text{Dom}(f) := \{x \in \mathbb{R}\} \quad \text{and} \quad \text{Ran}(f) := \{y \in \mathbb{R} | y > 0\}$$

From the graph of exponential function we see that this is a one-to-one function. It follows that

$f(x) = a^x$ has an inverse function called Logarithmic function that denoted as $y = \log_a x$.

a is called the base of the function..



The two most common bases for the Logarithmic Function are 10 and e.

Logarithms with a base of 10 are called common logarithms: $\log_{10} x = \log x$

Logarithms with a base of e are called natural logarithms: $\log_e x = \ln x$

Properties of Logarithms:

1. $\log_a 1 = 0$ 2. $\log_a a = 1$ 3. $\log_a a^x = x$ 4. $a^{\log_a x} = x$

5. $\log_a b = \frac{\ln b}{\ln a} = \frac{\log b}{\log a}$ 6. $\log_a x^b = b \log_a x$

7. $\log_a (uv) = \log_a u + \log_a v$ 8. $\log_a \left(\frac{u}{v}\right) = \log_a u - \log_a v$

Example 1

Find the exact value of each expression:

a) $\log_2 6 - \log_2 15 + \log_2 20 =$

b) $\log \sqrt{10} =$

c) Convert to natural exponential function $-2(3^x) =$

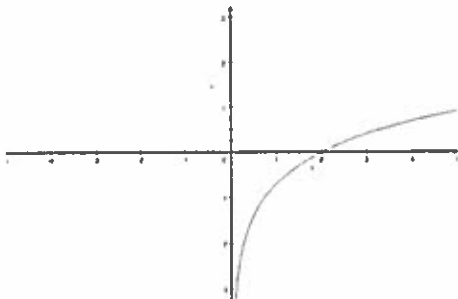
d) For all $x \in \mathbb{R}, x > 0$, $2^{\log_2 x} =$

e) For all $x \in \mathbb{R}$, $\log_4(4^x) =$

f) Solve the following inequality: $\ln \frac{x-1}{2} > 0$

I.

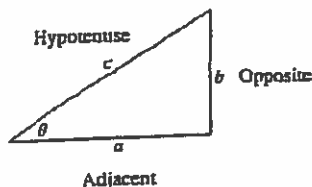
II.



2.3. Trigonometric Functions.

Degree measure, traditionally used to measure angles. One full revolution is 360degrees. There is another way of measuring angles which is more natural and more useful for methods used in calculus. This method is measuring angles in 'radians'. The idea is measuring angles by considering rotations. One full revolution in radian is 2π radians. The circumference of a circle of radius 1 is 2π . Thus 2π radians=360 degrees. When we use radians, angles are measured with length of the corresponding arcs on the unit circle.

The trigonometric functions $\sin \theta = \frac{b}{c} = \frac{\text{opposite}}{\text{hypotenuse}}$; $\cos \theta = \frac{a}{c} = \frac{\text{adjacent}}{\text{hypotenuse}}$



A disadvantage of this definition is that it makes sense only if θ lies between 0 and $\frac{\pi}{2}$

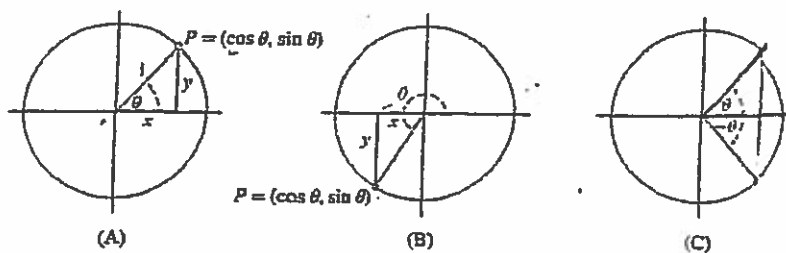
However, sine and cosine can be defined for all angles in terms of unit circle. Let $P(x,y)$ be the point on the unit circle corresponding to the angle θ and define

$$\cos \theta = x - \text{coordinate of } P \quad \sin \theta = y - \text{coordinate of } P$$

On the circle of radius r the point corresponding to the angle θ has coordinates $(r \cos \theta, r \sin \theta)$

Furthermore,

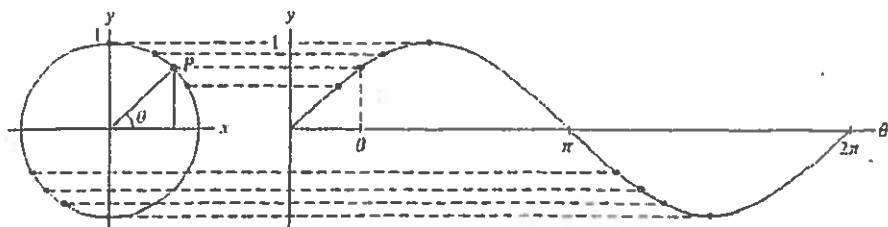
sine is an odd function $\sin(-\theta) = -\sin \theta$ and cosine is an even function $\cos(-\theta) = \cos \theta$.



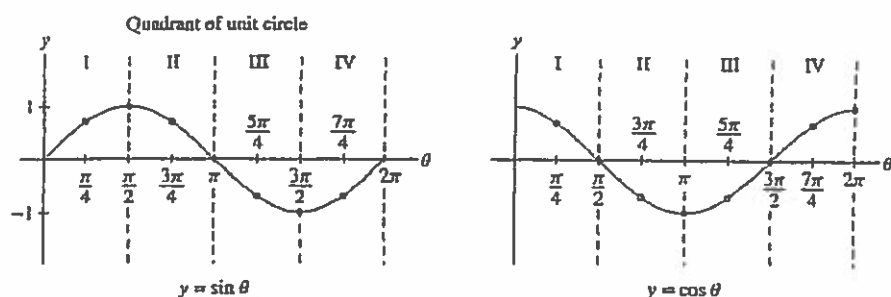
Although, we use calculator to evaluate sine and cosine for general angles, the standard values listed in the table appear often and should be memorized.

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{3}}{2}$	-1

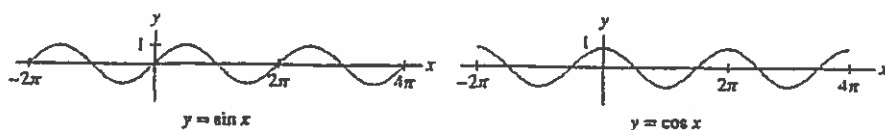
The graph of $y = \sin x$ is the familiar sine wave, generated by the y -coordinate of the point $P = (\cos \theta, \sin \theta)$ moving around the unit circle.



The graph of $y = \cos x$ has the same shape, but is shifted to the left by $\frac{\pi}{2}$ units.



A function $f(x)$ is called periodic with period T if $f(x+T) = f(x)$, where T is the smallest positive number with this property. The sine and cosine functions are periodic with period 2π .



There are four other standard trigonometric functions:

$$\tan x = \frac{\sin x}{\cos x} \qquad \cot x = \frac{\cos x}{\sin x}$$

$$\sec x = \frac{1}{\cos x} \qquad \csc x = \frac{1}{\sin x}$$

2.4. Trigonometric Identities

Pythagorean identities: $\sin^2 x + \cos^2 x = 1$ $\tan^2 x + 1 = \sec^2 x$ $\cot^2 x + 1 = \csc^2 x$

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$$

Sum and Difference identities: $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$

$$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \pm \tan x \tan y}$$

$$\sin 2x = 2 \sin x \cos x$$

Double angle identities: $\cos 2x = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$

$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$$

Power reducing identities: $\sin^2\left(\frac{x}{2}\right) = \frac{1 - \cos x}{2}$ $\cos^2\left(\frac{x}{2}\right) = \frac{1 + \cos x}{2}$

2.5. Inverse Trigonometric Functions.

Trigonometric Functions are periodic, not one-to-one functions. In order to define inverses for all trigonometric functions we have to restrict their domains and work only with *principal values* of the functions.

The following restrictions on the domains of the functions make the principal values of the functions invertible:

$$y = \sin x, \quad x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$y = \cos x, \quad x \in [0, \pi]$$

$$y = \tan x, \quad x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$y = \cot x, \quad x \in (0, \pi)$$

The domain of trigonometric function becomes the range of corresponding inverse function :

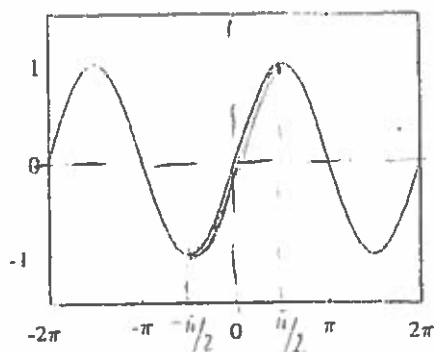
$$y = \sin^{-1} x = \arcsin x, \quad y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$y = \cos^{-1} x = \arccos x, \quad y \in [0, \pi]$$

$$y = \tan^{-1} x = \arctan x, \quad y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$y = \cot^{-1} x = \operatorname{arccot} x, \quad y \in (0, \pi)$$

Principal Values of Trigonometric Functions.

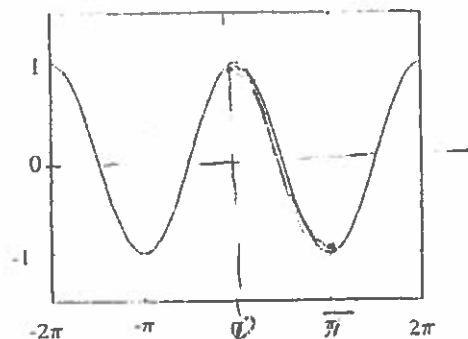


$$f(x) = \sin(x)$$

$$\text{Domain: } (-\infty, \infty)$$

$$\text{Range: } [-1, 1]$$

$$\text{Period: } 2\pi$$

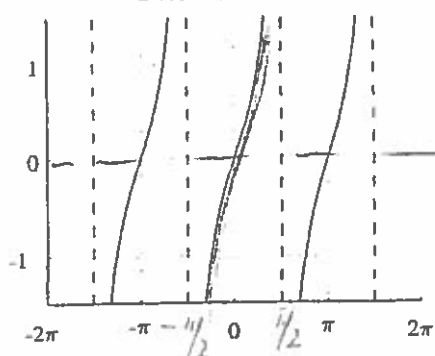


$$f(x) = \cos(x)$$

$$\text{Domain: } (-\infty, \infty)$$

$$\text{Range: } [-1, 1]$$

$$\text{Period: } 2\pi$$

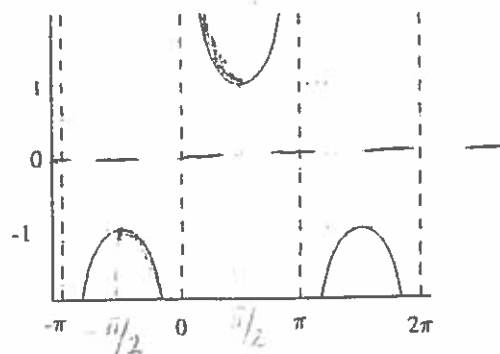


$$f(x) = \tan(x)$$

$$\text{Domain: } ((k - \frac{1}{2})\pi, (k + \frac{1}{2})\pi)$$

$$\text{Range: } (-\infty, \infty)$$

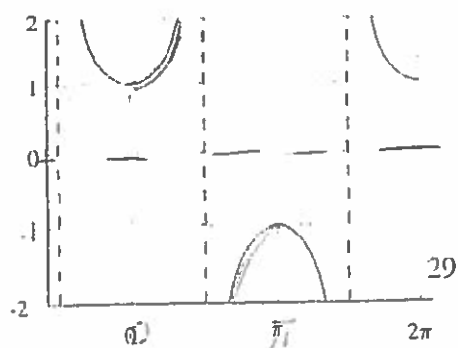
$$\text{Period: } \pi$$



$$f(x) = \csc(x) = \frac{1}{\sin(x)}$$

$$\text{Domain: } ((k - 1)\pi, k\pi)$$

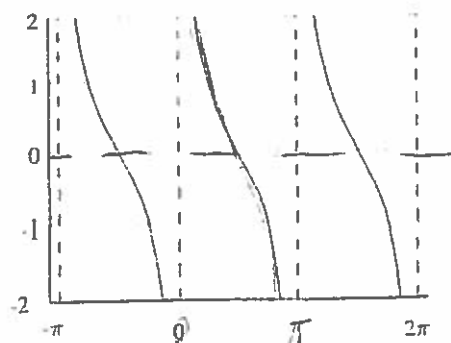
$$\text{Range: } (-\infty, -1] \cup [1, \infty)$$



$$f(x) = \sec(x) = \frac{1}{\cos(x)}$$

$$\text{Domain: } ((k - \frac{1}{2})\pi, (k + \frac{1}{2})\pi)$$

$$\text{Range: } (-\infty, -1] \cup [1, \infty)$$

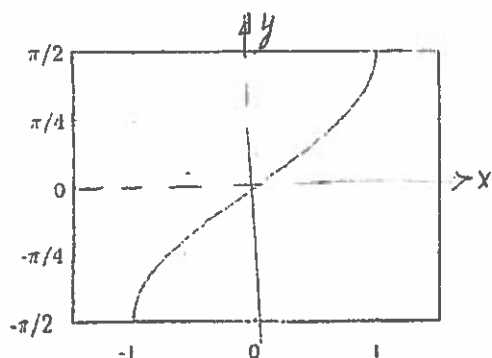


$$f(x) = \cot(x) = \frac{1}{\tan(x)}$$

$$\text{Domain: } ((k - 1)\pi, k\pi)$$

$$\text{Range: } (-\infty, \infty)$$

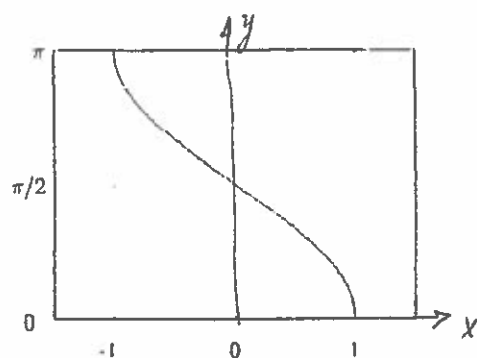
Graphs of Inverse Trigonometric Functions.



$$f(x) = \sin^{-1}(x) \equiv \arcsin(x)$$

$$\text{Domain: } [-1, 1]$$

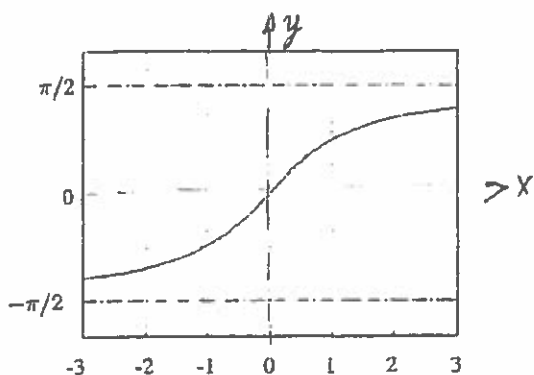
$$\text{Range: } \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$



$$f(x) = \cos^{-1}(x) \equiv \arccos(x)$$

$$\text{Domain: } [-1, 1]$$

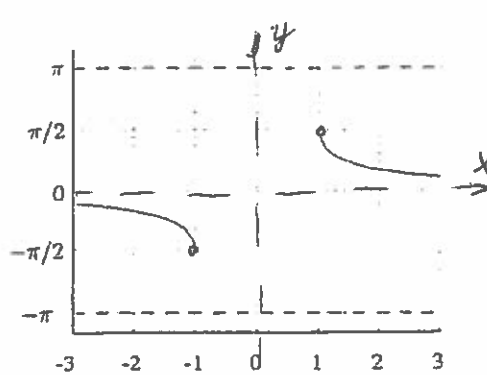
$$\text{Range: } [0, \pi]$$



$$f(x) = \tan^{-1}(x) \equiv \arctan(x)$$

$$\text{Domain: } (-\infty, \infty)$$

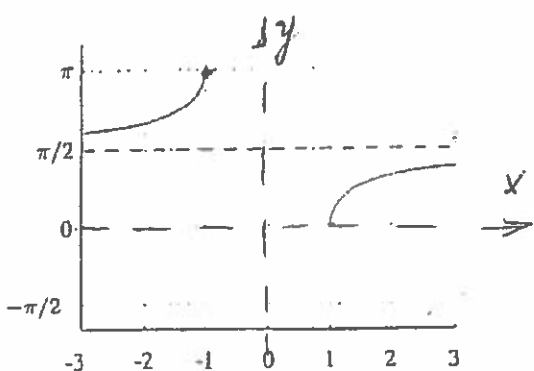
$$\text{Range: } \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$



$$f(x) = \csc^{-1}(x) \equiv \operatorname{arccsc}(x)$$

$$\text{Domain: } (-\infty, -1] \cup [1, \infty)$$

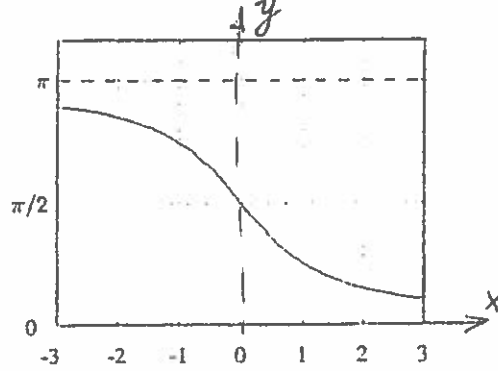
$$\text{Range: } \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], y \neq 0$$



$$f(x) = \sec^{-1}(x) \equiv \operatorname{arcsec}(x)$$

$$\text{Domain: } (-\infty, -1] \cup [1, \infty)$$

$$\text{Range: } [0, \pi], y \neq \pi/2$$



$$f(x) = \cot^{-1}(x) \equiv \operatorname{arccot}(x)$$

$$\text{Domain: } (-\infty, \infty)$$

$$\text{Range: } (0, \pi)$$

Example:

Find **all** solutions to the given trigonometric equation : $4\sin^2 x = 1$

Example:

Find **all** solutions to the given trigonometric equation: $\cos x = \frac{\sqrt{3}}{2}$

Example:

Find **all** solutions to the given trigonometric equation: $\sqrt{3} \cot 3x + 1 = 0$

